

Kai Li

Data Science, Applied Machine Learning, and Reinforcement Learning

Stony Brook, NY | kai.li@stonybrook.edu | garylikai.github.io | [LinkedIn](#) | [GitHub](#)

SUMMARY

Statistics Ph.D. candidate with peer-reviewed research in reinforcement learning, optimization, simulation, and statistical machine learning. Builds reproducible computational studies in Python and R, including custom environments, controlled model comparisons, replicated experiments, held-out evaluation, and statistical analysis. First author of a journal article and co-first author of an ICML workshop paper; seeking 2027 data science, applied scientist, machine learning, or reinforcement learning roles.

TECHNICAL SKILLS

Programming: Python, R, SQL. **Libraries and frameworks:** scikit-learn, TensorFlow/Keras, Gymnasium, Stable-Baselines3.

ML and RL: supervised learning, neural networks, Monte Carlo control, SARSA, Q-learning, deep Q-networks, proximal policy optimization, Markov decision processes.

Statistics and computing: regression, experimental design, time series, simulation, Monte Carlo methods, model evaluation, optimization, simulated annealing, data visualization.

Tools: Git, GitHub, Jupyter, RStudio, Visual Studio Code, L^AT_EX.

SELECTED RESEARCH AND APPLIED MACHINE LEARNING EXPERIENCE

Reinforcement Learning Researcher — Blackjack Strategy

September 2025–Present

Collaborative research project, Stony Brook University

Stony Brook, NY

- **Publication study (March 2026–Present):** Compare six tabular and deep reinforcement learning algorithms: on-policy Monte Carlo, off-policy Monte Carlo, SARSA, Q-learning, deep Q-networks, and proximal policy optimization.
- Study algorithm behavior as the legal action space expands from Hit/Stand to settings that also include Double and Split, using a common Gymnasium-compatible simulator with state-dependent action handling.
- Use replicated training, held-out seeds, learning curves, measures of policy agreement, and matched decision-level comparisons; preparing a manuscript as a co-first author.
- **Pilot study (September–December 2025):** Compared Monte Carlo control and proximal policy optimization and identified confounding from action-space differences, split-hand credit assignment, reward variance, and deck assumptions.

Researcher and First Author — Cumulative Tic-Tac-Toe

July 2024–March 2026

Department of Applied Mathematics and Statistics, Stony Brook University

Stony Brook, NY

- Formulated a cumulative-scoring game as a Markov decision process and built a temporal-difference learning framework for empirical policy evaluation.
- Combined exhaustive game analysis and combinatorial game theory to establish that the game is a draw under optimal play.
- Designed a domain-informed evaluation function that reduced average training episodes by approximately 23.1% versus random initialization.
- Published the completed work as a first-author article in *Stats*; maintained a public research code repository tied to the published computational record.

Researcher and Co-First Author — RL-QESA

December 2024–July 2025

Collaborative research project, Stony Brook University

Stony Brook, NY

- Modeled temperature control for simulated annealing as a block-level Markov decision process in which cooling is gated by a quasi-equilibrium condition.
- Used a transformer-based policy trained with proximal policy optimization to adapt cooling decisions from block-level energy and acceptance statistics.
- Helped establish conditions retaining classical convergence guarantees for simulated annealing and evaluated performance on Rosenbrock and Lennard–Jones optimization tasks against classical and neural baselines.
- Published the work at the AI for Math Workshop at ICML 2025 as a co-first author.

Graduate Research Assistant

May 2025–August 2025

Institute for Advanced Computational Science, Stony Brook University

Stony Brook, NY

- Assessed reinforcement learning methods for climate model parameterization and multi-agent adaptation to sea-level rise through technical literature review and mathematical formulation.
- Compared eight model-free algorithms on sample efficiency, stability, hyperparameter sensitivity, and reproducibility; translated a published adaptation model into Markov decision process components and Bellman equations.
- Produced two technical presentations and an implementation-oriented outline for a future multi-agent advantage actor–critic simulation study.

SELECTED DATA SCIENCE PROJECTS

Time Series Forecasting of Store Sales

February–April 2022

- Implemented ARIMA, recurrent neural network, LSTM, and GRU forecasting workflows in R and Keras and compared predictive performance using root mean squared error.

Understanding Flight Delays

August–December 2021

- Prepared 2012–2019 airline data and compared ridge regression, multilayer perceptrons, and k -nearest neighbors; used statistical tests to assess model performance and analyzed seasonal and airline-specific delay patterns.

PMLi: Statistical Methods for Partially Matched Samples

March–May 2021

- Developed and documented an R package implementing methods for pooling p -values from mixtures of independent and matched observations, missing data, and exceptional cases.

ADDITIONAL EXPERIENCE

Graduate Instructor and Teaching Assistant

August 2022–Present

Department of Applied Mathematics and Statistics, Stony Brook University

Stony Brook, NY

- Taught or supported nine undergraduate and graduate sections in statistical computing, mathematical statistics, time series, and statistical learning, serving 320 students; earned five teaching distinctions, including two Teachers Rated Excellent Educators by their Students (TREES) recognitions.

EDUCATION

Stony Brook University, Stony Brook, NY

Expected May 2027

Ph.D. in Applied Mathematics and Statistics, Statistics Track; GPA: 4.00/4.00

Stony Brook University, Stony Brook, NY

December 2022

M.S. in Applied Mathematics and Statistics; GPA: 3.95/4.00. Completed requirements for both Statistics and Operations Research tracks.

Stony Brook University, Stony Brook, NY

2022

Advanced Graduate Certificates in Operations Research and Data and Computational Science

The Ohio State University, Columbus, OH

May 2020

B.S. in Mathematics, Theoretical Track; minors in Computer Information Science and Economics

SELECTED RESEARCH OUTPUTS

- Li, K. and Zhu, W. “Combinatorial Game Theory and Reinforcement Learning in Cumulative Tic-Tac-Toe via Evaluation Functions.” *Stats*, 9(2), Article 28, 2026. doi:10.3390/stats9020028.
- Xu, R., Li, K., Wang, H., Kementzidis, G., Zhu, W., and Deng, Y. “RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing.” *2nd AI for Math Workshop at ICML 2025*, 2025. [OpenReview](#). Co-first author.